

FILTER DESIGN TEST CASE

Coupled Resonator Filter - Band 41 (2500 MHz TDD) for Satellite L-Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0009
Revision:	Rev A.1
Author:	Dr. Ahmed Al-Farsi
Reviewer:	Dr. Mei-Ling Chen
Design Center:	Andover HQ Lab
Created Date:	2025-03-26
Last Modified:	2025-05-04
Status:	NEEDS REVIEW
Priority:	P3 - Medium

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Coupled Resonator Filter filter targeting Band 41 (2500 MHz TDD) for Satellite L-Band Filter. The filter is designed on 42° YX LiTaO3 substrate with a target center frequency of 3158.3 MHz and bandwidth of 340.2 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Satellite L-Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Coupled Resonator Filter
- Frequency Band: Band 41 (2500 MHz TDD)
- Application: Satellite L-Band Filter
- Substrate: 42° YX LiTaO3
- Package: QFN (Quad Flat No-Lead)
- Design Tool: CST Studio Suite 2024
- Design Center: Andover HQ Lab

This specification applies to all variants of the Coupled Resonator Filter design targeting Band 41 (2500 MHz TDD). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3158.3 MHz
Bandwidth (3 dB):	340.2 MHz
Insertion Loss Target:	<= 3.3 dB
Return Loss Target:	>= 15.0 dB
Out-of-Band Rejection:	>= 42.0 dB
Isolation (Duplexer):	>= 21.0 dB
Q Factor:	10997
Temperature Coefficient:	-29.8 ppm/°C
Die Size:	2.6 x 1.0 mm
Wafer Size:	8-inch
Number of Resonators:	10
Electrode Material:	Pt/Al

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

- 1. Open CST Studio Suite 2024 and load the Coupled Resonator Filter template project.
- 2. Define the substrate stack: 42° YX LiTaO3 with specified layer thicknesses.
- 3. Set design targets: center freq 3158.3 MHz, BW 340.2 MHz.
- 4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
- 5. Optimize resonator geometry using gradient descent (target IL < 3.3 dB).
- 6. Verify coupling coefficients between resonator stages.
- 7. Run 3D FEM simulation to account for package parasitics (QFN (Quad Flat No-Lead)).
- 8. Export GDS-II layout for mask fabrication.
- 9. Fabricate prototype wafers (8-inch, 42° YX LiTaO3).
- 10. Perform on-wafer probing using Keysight N9041B UXA.
- 11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 9475 MHz.
- 12. Compare measured results against simulation and specification limits.
- 13. Perform temperature cycling (-40°C to +85°C) and re-measure.
- 14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.3 dB
- Return loss in passband shall be >= 15.0 dB
- Out-of-band rejection at +/- 510 MHz offset >= 42.0 dB
- Group delay variation across passband shall be <= 6 ns
- Temperature drift over -40°C to +85°C shall be <= 11.8 MHz
- Power handling shall be >= +34 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 92%

6. TEST RESULTS

Measurement Date: 2025-05-04
Devices Tested: 18
Insertion Loss (meas): 2.97 dB
Return Loss (meas): 16.6 dB
Rejection (meas): 44.7 dB
Isolation (meas): 25.8 dB
Q Factor (meas): 10997
Group Delay Variation: 1.4 ns
Power Handling: +29 dBm
Die Yield: 91.9%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The Coupled Resonator Filter design demonstrated below-target performance for Band 41 (2500 MHz TDD).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Simulation-to-measurement correlation was within 3% for all key parameters.
- e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

Author:	Dr. Ahmed Al-Farsi	Date:	2025-05-04
Reviewer:	Dr. Mei-Ling Chen	Date:	2025-05-08
Approver:	Dr. Yuko Ishida	Date:	2025-05-15

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